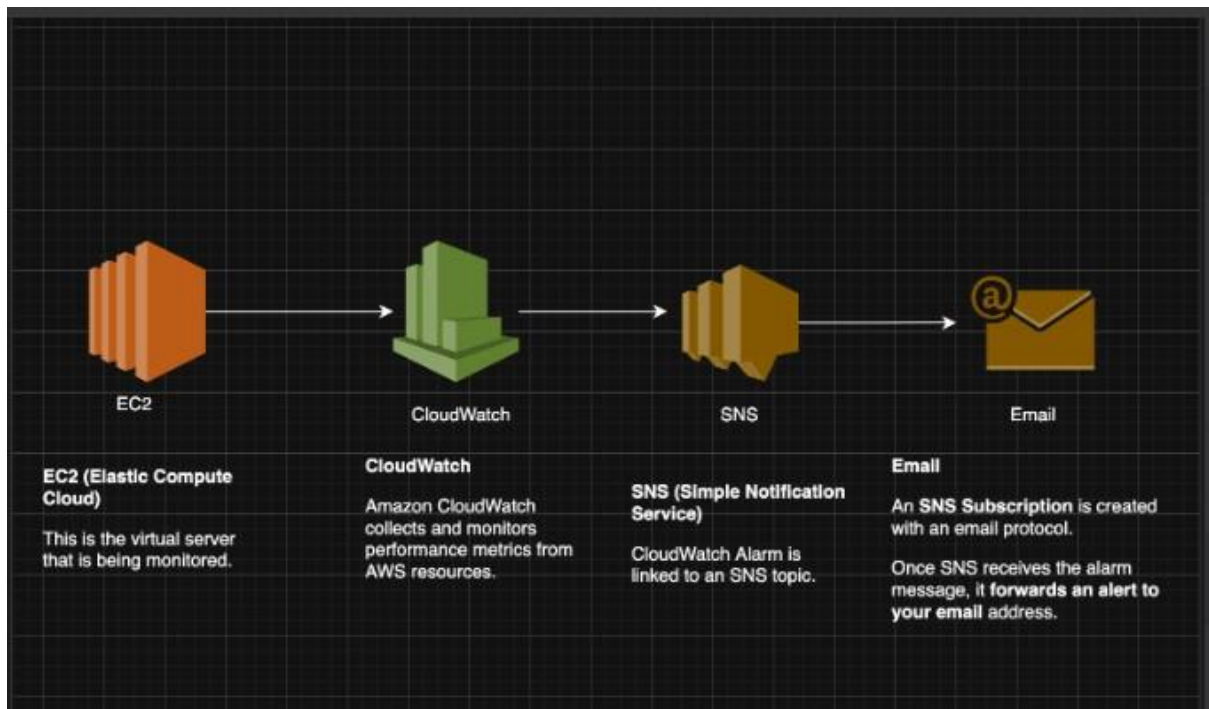
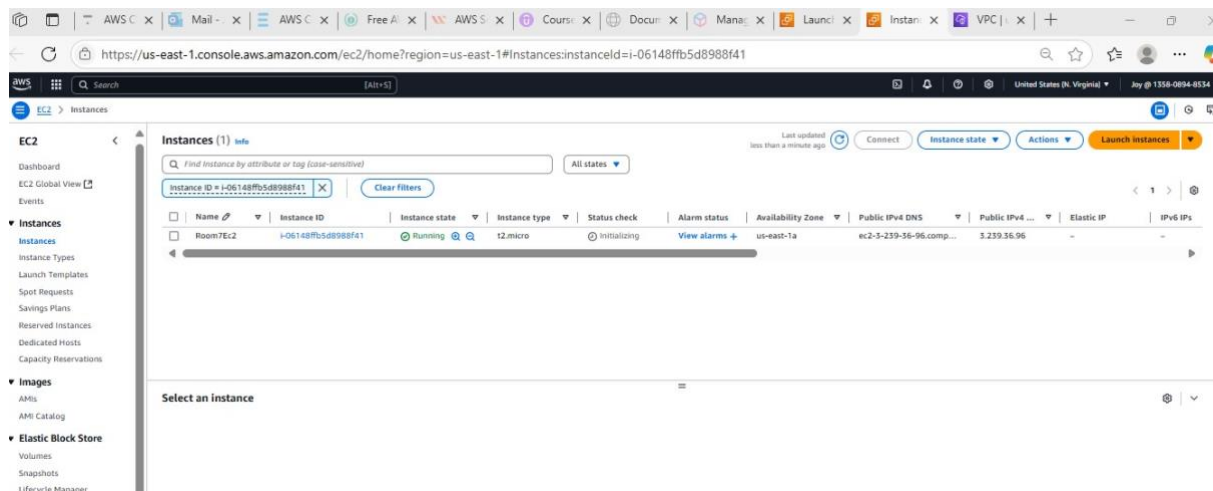


TOPIC: Set Up Monitoring and Alerting for EC2 Instance Health Using Amazon CloudWatch

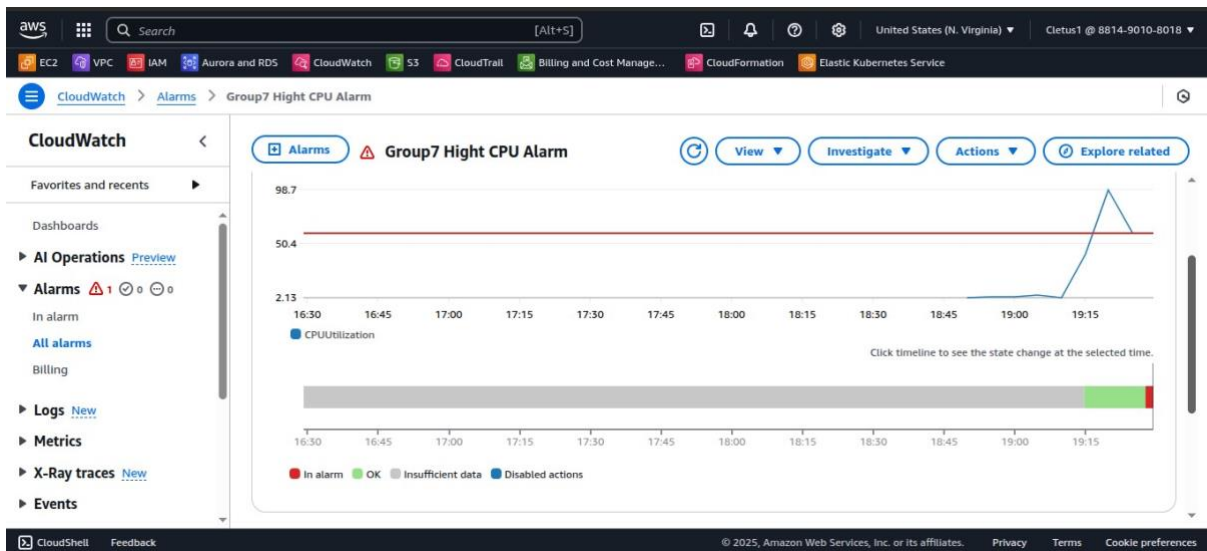
TASK 1: Architecture Diagram



TASK 2: We created a EC2 instance

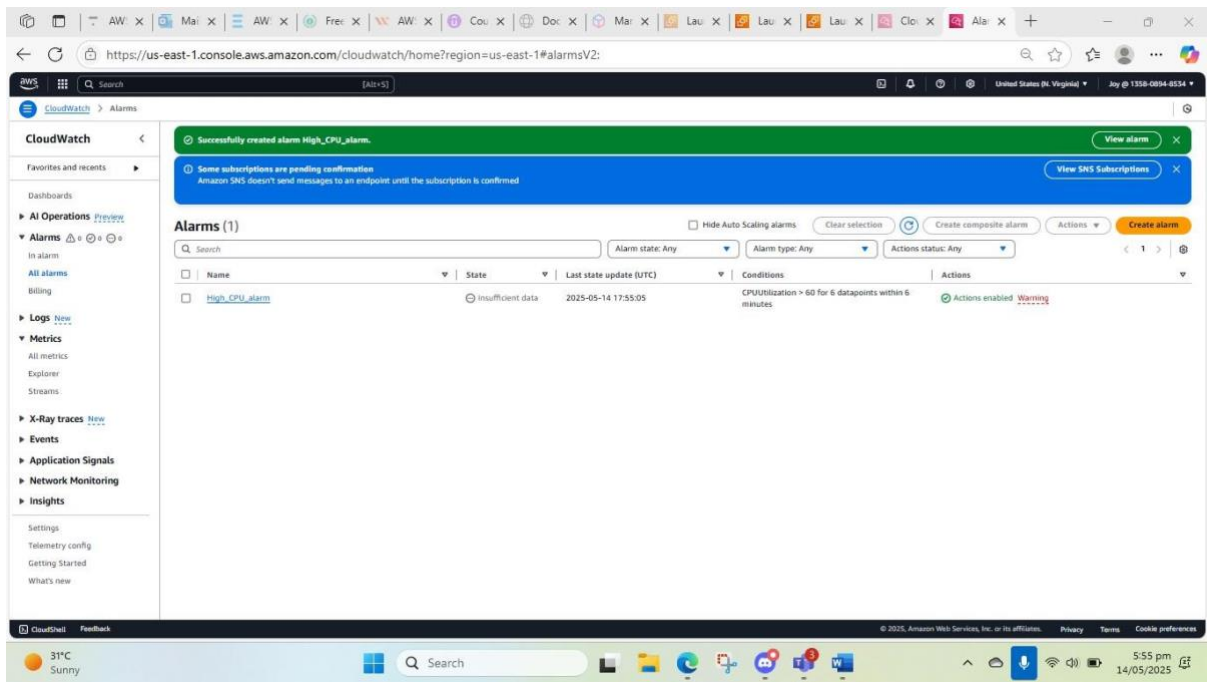


TASK 2: Screenshot of CloudWatch Alarms

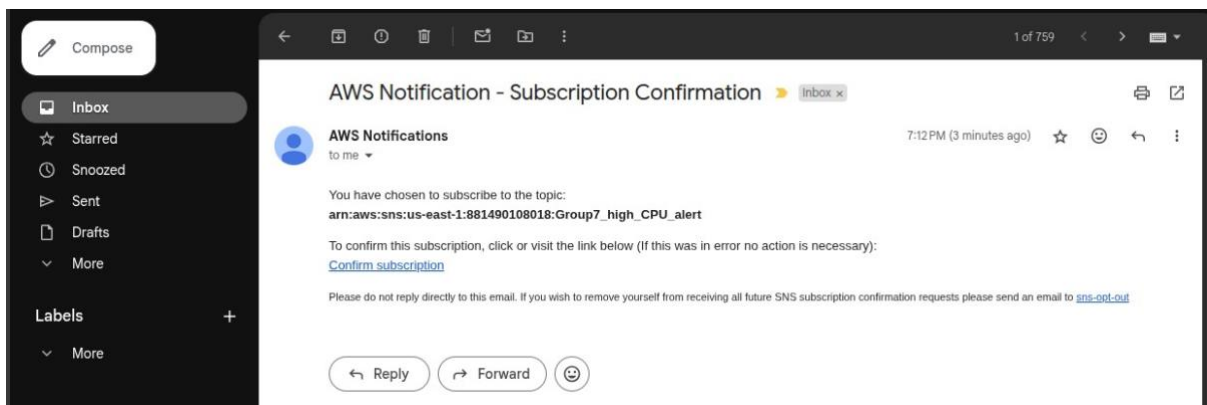


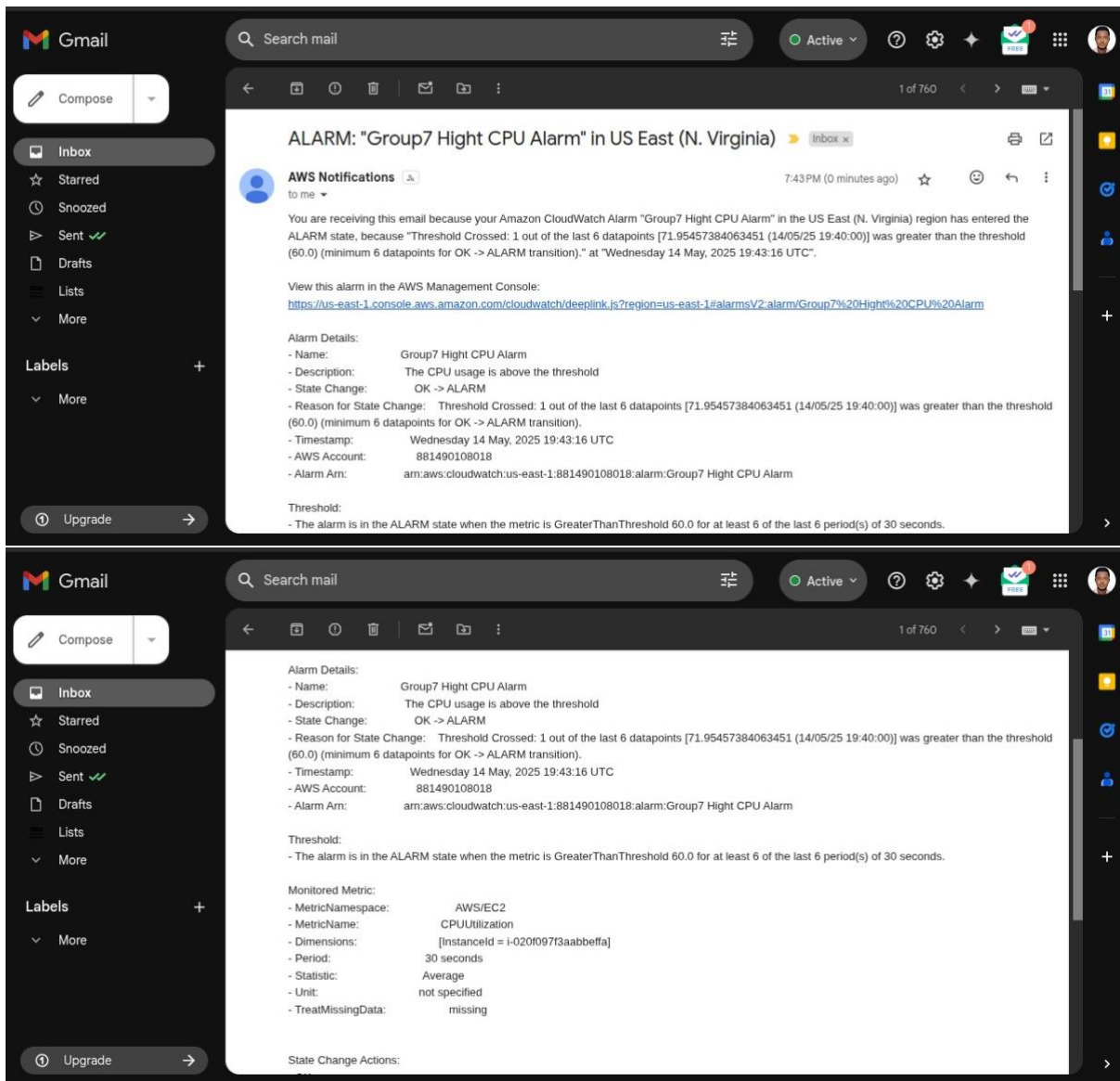
Created the alarm

The screenshot shows the AWS CloudWatch console for the 'Alarms (1)' list. The list contains one alarm named 'High_CPU_alarm' with a state of 'Insufficient data'. The alarm conditions are 'CPUUtilization > 60 for 6 datapoints within 6 minutes'. The actions are 'Actions enabled' with a warning icon. The top of the page shows a success message: 'Successfully created alarm High_CPU_alarm.' and a notification about pending SNS subscriptions. The left sidebar shows the navigation menu with 'Alarms' selected. The top navigation bar includes the AWS logo, search bar, and account information.



TASK 3: SNS Email Alert Screenshot





Using lac to achieve same results

```
aws
[Alt+S]
United States (N. Virginia)
Cletus1 @ 8814-9010-8018

EC2 VPC IAM Aurora and RDS CloudWatch S3 CloudTrail Billing and Cost Manage... CloudFormation Elastic Kubernetes Service

[ec2-user@ip-172-31-0-123 ~]$ sudo yum install -y stress
stress --cpu 2 --timeout 600
Last metadata expiration check: 0:12:25 ago on Wed May 14 19:04:42 2025.
Dependencies resolved.

=====
Package                Architecture      Version           Repository        Size
=====
Installing:
stress                 x86_64            1.0.7-2.amzn2023.0.1  amazonlinux       34 k
=====
Transaction Summary
=====
Install 1 Package

Total download size: 34 k
Installed size: 68 k
Downloading Packages:
stress-1.0.7-2.amzn2023.0.1.x86_64.rpm                                907 kB/s | 34 kB  00:00
-----
Total                                                                481 kB/s | 34 kB  00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 1/1
  Installing     : stress-1.0.7-2.amzn2023.0.1.x86_64 1/1
=====
```

Monitoring Report

1. What Did We Learned About Monitoring in AWS?

Monitoring in AWS helps track system health, performance, and security. Key tools include:

- Amazon CloudWatch for real-time metrics and automated alerts.
- Amazon SNS for notifications when thresholds are breached.
- EC2 Monitoring to observe CPU, memory, and network usage. This ensures proactive issue detection to maintain system reliability.

2. Scaling the Solution for Production

To handle production workloads efficiently:

- Enable Auto Scaling for EC2 instances to adjust capacity based on traffic.
- Use AWS Lambda to automate responses (e.g., restarting instances on failure).
- Leverage CloudWatch Logs for deeper insights and troubleshooting.
- Integrate AWS Systems Manager for centralized management.

3. Additional Metrics to Monitor

Beyond CPU utilization, we will consider:

- Memory Usage: Prevent resource exhaustion.
- Disk I/O: Detect storage bottlenecks.
- Network Traffic: Prevent security threats.

GROUP MEMBERS

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